

Scott Nguyen

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EDUCATION

Master of Science in Electrical Engineering Space Systems Engineering University of New Mexico	Fall 2026 GPA: 3.95/4.00
Master of Science in Aerospace Engineering University of Illinois Urbana-Champaign	Fall 2024
Bachelor of Science in Aerospace Engineering Iowa State University	Spring 2022

SKILLS

Programming Languages: MATLAB, Python, C/C++, Ruby
Frameworks / Libraries / Tools: NumPy, SciPy, Matplotlib, Astropy, Poliastro, bpy, git
Applications: Simulink, Blender, ANSYS HFSS, FEKO, TICRA

WORK EXPERIENCE

Student Co-Op, Electronics for Contested Space Group <i>MIT Lincoln Laboratory</i>	September 2025 – January 2026
- Developed an optical space domain awareness analysis tool that propagates TLE-based RSO orbits from user-defined sensor locations and applies illumination and elevation constraints to identify observable windows and pointing solutions. - Modeled RSO detectability and tracking performance for optical space domain awareness using photometric, kinematic, and sensor-noise models to assess detection limits and track duration. - Designed a communication architecture with radio-frequency hardware and simulation to support lost-satellite reacquisition.	
Guidance, Navigation & Controls Engineer Intern <i>Blue Canyon Technologies</i>	May 2025 – August 2025
- Executed post-environment testing of IMU, star trackers, reaction wheels, torque rods, and sun sensors. - Diagnosed Solar Array Drive Assembly (SADA) timing latency and automated COSMOS (Ruby) validation.	
Guidance, Navigation & Controls Engineer Intern <i>Blue Origin</i>	January 2025 – April 2025
- Designed a hybrid <i>Active Disturbance Rejection Control (ADRC)</i> + <i>Sliding Mode Control (SMC)</i> controller; used an extended state observer and a sliding manifold for robust tracking under uncertainty and actuator limits. - Injected time-domain disturbances and conducted a trade study comparing the hybrid controller to <i>PID</i> and <i>LQR</i> on LTI systems; benchmarked rise time, settling time, and overshoot. - Integrated flight software into <i>Simulink</i> via <i>C S-functions</i> to run full closed-loop simulations.	
Guidance, Navigation & Controls Engineer Intern <i>Varda Space Industries</i>	May 2024 – August 2024
- Conducted trade studies to optimize gravity models for mission requirements and select the best filter (<i>Extended Kalman Filter (EKF)</i> vs. <i>Unscented Kalman Filter (UKF)</i>) - Created Monte Carlo simulations to perform flight safety analysis and develop reentry criteria for capsule reentry - Implemented an <i>EKF</i> for state estimation, optimizing ground station data timing to minimize residuals and enable precise delta-v planning - Implemented unit testing for the code base and introduced CI/CD pipelines using Bamboo	
Guidance, Navigation & Controls Engineer Intern <i>Space Dynamics Laboratory</i>	May 2023 – August 2023
- Developed angles-only orbit determination pipelines using UKF, range iteration, and least-squares methods for SSA tracking and characterization. - Performed Monte Carlo observability analysis to identify challenging geometries and improve tracking of closely spaced RSOs. - Implemented and unit-tested a Lambert-based IOD solver to support track initialization and catalog maintenance.	

RESEARCH PROJECTS

Optical Orbit Determination and Tracking
- Built an end-to-end orbit determination pipeline from ground-based optical line-of-sight measurements, including time/frames handling and measurement conditioning - Applied classical initial orbit determination (Laplace) to recover an ECI state from sparse measurements; validated against truth propagation and evaluated sensitivity to realistic measurement noise via Monte Carlo error statistics
Astrodynamics and Optimization for Mission Design (Earth–Mars Transfer)
- Designed and analyzed interplanetary transfers using ephemeris-driven Lambert targeting and patched-conic ΔV estimates to assess launch-window trade spaces. - Combined grid-based porkchop mapping with constrained nonlinear optimization (SLSQP) to identify low-ΔV solutions under time-of-flight constraints; produced 3D trajectory visualizations for verification and communication